

LAB 3 : TokTickIT Authenticated IT Ticketing System

Final submission report - authenticated users, IT Staff ticketing, Administrator safety, and release evidence

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Answer Part 1: Git Use with Engineering Workflow (10 Points)

Lab 3 was developed on issue-specific branches, reviewed through pull requests, integrated into lab3-staging, and released to main. The final release is PR #46 at commit 92fac36.

Evidence	Link
GitHub repository	HolyThiccDaddy/toktickit
GitHub project board	TokTickIT Individual Sprints
Issue #36 - authentication and migration	Issue #36
Issue #37 - requester authorization regression	Issue #37
Issue #38 - staff queue and ticket operations	Issue #38
Issue #39 - Administrator and release evidence	Issue #39
Final release PR #46	PR #46 - merged into main

Final release status: PR #46 was peer-approved and merged into main as 92fac36. The documentation follow-up in this report is submitted separately as PR #47.

Answer Part 1: Git Use with Engineering Workflow (continued)

Peer-review and release history

Stage	Pull request and result
Contract	PR #40 approved and merged as 40ca398.
Authentication and migration	PR #41 merged as b49c8b2 after credential and path-gate fixes.
Requester regression	PR #42 merged as 0ee0002.
Staff queue	PR #43 merged as c1cd708.
Administrator implementation	PR #44 approved after requested changes and merged as bd5872a.
Documentation follow-up	PR #45 approved and merged as e82ac21.
Final integration	PR #46 approved and merged into main as 92fac36.

The reviewer record identifies requested changes, responses, approval reviews, and merge commits. The final board and branch state are recorded in the source documents linked in Parts 2 and 3.

Answer Part 2: Spec DD (5 Points)

Approved engineering contract and traceability

Document	Purpose and evidence
Specification	docs/lab-03/specification.md - requirements, roles, business rules, data model, migration, and Definition of Done.
API contract	docs/lab-03/api-spec.md - named request/response schemas, status codes, authorization, and compatibility.
UI specification	docs/lab-03/ui-spec.md - screens, states, Zen Green tokens, accessibility, and responsive rules.
Test plan	docs/lab-03/tests.md - acceptance-criterion traceability and real execution evidence.

Contract-before-implementation evidence

PR #40 records the approved contract before implementation. It covers authentication, first-login password change, role and ownership authorization, Lab 2 migration, staff queue/detail, comments and notes, Administrator safeguards, responsive UI, and acceptance-criterion tests.

The contract is extended by the final release record rather than replaced: the merged main commit is 92fac36, and the final PDF is generated from the source tree at that release.

Key contract decisions

- Three authenticated roles: REQUESTER, IT_STAFF, and ADMINISTRATOR, with server-side authorization.
- The first-login password gate applies globally to protected APIs until the new password is saved.
- Lab 2 requester ownership and identifiers are preserved through migration and idempotent seed.
- IT Priority, complete status transitions, public comments, and staff-only Internal Notes use named schemas and tests.
- Administrator account safety prevents duplicate emails, self-deactivation, and removal of the last active Administrator.

Answer Part 3: Test DD and Traceability (10 Points)

Verification results from the merged Lab 3 release

Check	Executed result
Server regression	15 test files / 92 tests passed twice against the dedicated _test database.
Client regression	15 test files / 50 tests passed twice; the Create Ticket loading race is stabilized.
Server production build	TypeScript production build passed.
Client production build	TypeScript and Vite production build passed.
Playwright E2E	13 tests passed across desktop, tablet, and mobile projects.
Migration and seed	Seven migration/regression tests passed, including preserved Lab 2 rows, collision-safe IDs, pending credentials, and deterministic reseeding.
Repository hygiene	git diff --check passed for source documentation; local environment files and credentials remain ignored.

Acceptance-criterion traceability

Criteria	Evidence mapped in tests.md
AC-01 to AC-04	Login, first-login password gate, logout/expiry, CSRF, and role-aware session APIs and UI tests.
AC-05 to AC-06	Requester ownership, comments, resolution indication, attachment compatibility, and regression E2E.
AC-07 to AC-09	Shared queue, staff detail, assignment, IT Priority, status matrix, public comments, and Internal Notes.
AC-10 to AC-13	Administrator listing/filtering, create/edit/activation/reset, safety protections, migration, and forged-identity rejection.
AC-14 to AC-15	Responsive screenshots, visual checks, complete regression suites, production builds, and release integration.

Detailed test-file paths and command results are maintained in [docs/lab-03/tests.md](https://github.com/department-of-homeland-security/doc-examples/blob/main/docs/lab-03/tests.md).

Answer Part 4: AI Use with Reflection (5 Points)

LLM used: OpenAI Codex (GPT-5). The agent output was treated as a draft and accepted only after comparison with Lab_3_sheet.pdf, the approved contract, executable tests, real terminal evidence, and peer-review feedback.

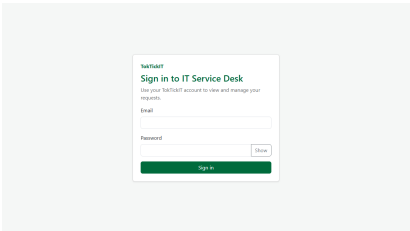
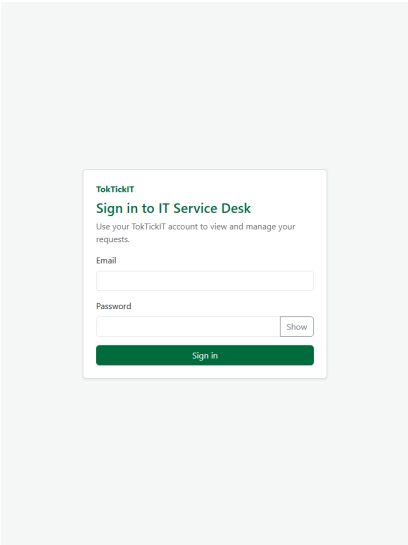
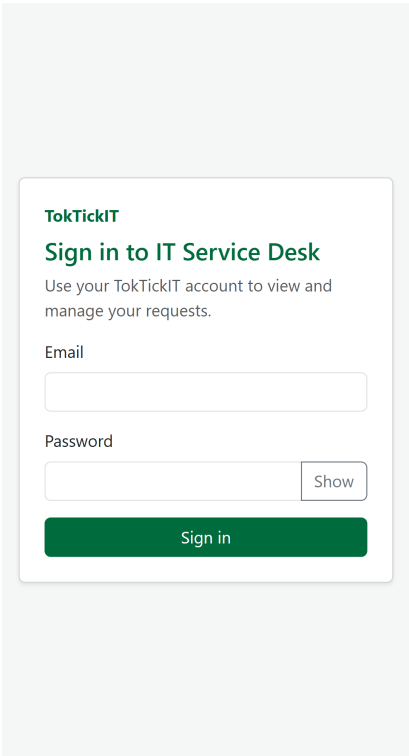
#	Selected prompt and accepted result
1	Convert the Lab 3 handout into a traceable product increment, issue plan, and Definition of Done.
2	Draft the engineering contract for authentication, role authorization, migration, queue/detail, comments/notes, and Administrator safety.
3	Implement the server foundation with sessions, password gate, CSRF, migration compatibility, and regression tests.
4	Implement requester authorization and communication while preserving Lab 2 ownership and attachment behavior.
5	Design Staff Queue, Ticket Detail, User Management states, responsive layouts, and safe failure feedback.
6	Map every acceptance criterion to unit, API, UI, security, migration, responsive, and E2E checks.
7	Recheck peer-review comments for contract, migration, authorization, concurrency, filtering, and E2E evidence gaps.
8	Audit release evidence for source/test-count consistency and final-main readiness.

My Reflection

Codex was most useful for turning the handout into a traceable contract, finding mismatches between requirements and tests, and generating focused regression cases. Human decisions and peer review remained authoritative: authorization, migration safety, password handling, and status transitions were accepted only after executable tests and real evidence passed. Suggestions that weakened server-side checks, exposed credentials, relied on hidden UI controls, or added excluded features were changed or rejected. The final release was rechecked from merged main before this report was generated.

Answer Part 5: Working Login and Password Change UI (5 Points)

The authenticated session boundary replaces the Lab 2 Development Requester selector. The UI covers valid and invalid login, inactive-account handling, first-login password change, role-aware navigation, logout, and direct-access protection after logout.

Authentication and first-login password-change evidence		
		
Desktop	Tablet	Mobile

Real Playwright captures from the merged Lab 3 release. Desktop, tablet, and mobile views show the same Zen Green form hierarchy without clipping or horizontal overflow.

Supporting tests: server/tests/lab-03/auth.api.test.ts, authorization.api.test.ts; client/tests/lab-03/Login.test.tsx and ChangePassword.test.tsx; e2e/lab-03/authentication.spec.ts.

Answer Part 6: Working IT Staff Ticket Queue UI (5 Points)

IT Staff and Administrators share a searchable queue with status, IT Priority, assignee, category, sorting, pagination, realistic empty/no-results states, retryable failure feedback, and a responsive card layout.

IT Staff Ticket Queue responsive evidence

TokTicketIT

Ticket Queue

Alex Staff
IT_STAFF

Sign out

Ticket Queue

Shared queue for IT Staff.

IT_STAFF

Search queue

Ticket, summary, requester, or ca

Status

All statuses

IT Priority

All priorities

Assignee ID

Any assignee

Category

All categories

Sort by

Updated date

Sort direction

Descending

Results per page

20

Clear

Ticket No.	Updated	Status	Summary	Requester	IT Priority	Assignee	Actions
TKT-2026-000001	9/17/2026, 4:48:58 PM	NEW	Issue 39 visual desktop	Sarah Johnson	HIGH	Unassigned	Open

1 result · Page 1 of 1

TokTicketIT

Ticket Queue

Alex Staff
IT_STAFF

Sign out

Ticket Queue

Shared queue for IT Staff.

IT_STAFF

Search queue

Ticket, summary, requester, or category

Status

All statuses

IT Priority

All priorities

Assignee ID

Any assignee

Category

All categories

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Results per page

20

Clear

Ticket No.	Updated	Status	Summary	Requester	IT Priority	Assignee	Actions
TKT-2026-000001	9/17/2026, 4:48:58 PM	NEW	Issue 39 visual tablet	Sarah Johnson	HIGH	Unassigned	Open

1 result · Page 1 of 1

Desktop

TokTicketIT

Ticket Queue

Alex Staff
IT_STAFF

Sign out

Ticket Queue

Shared queue for IT Staff.

IT_STAFF

Search queue

Ticket, summary, requester, or category

Status

All statuses

IT Priority

All priorities

Assignee ID

Any assignee

Category

All categories

Sort by

Updated date

Sort direction

Descending

Results per page

20

Clear

TKT-2026-000002

NEW

Issue 39 visual mobile

Requester: Sarah Johnson

HIGH Unassigned

Updated 9/17/2026, 4:49:14 PM

Open ticket

TKT-2026-000001

NEW

Issue 39 visual tablet

Requester: Sarah Johnson

HIGH Unassigned

Updated 9/17/2026, 4:48:58 PM

Open ticket

2 results · Page 1 of 1

Previous

1

Next

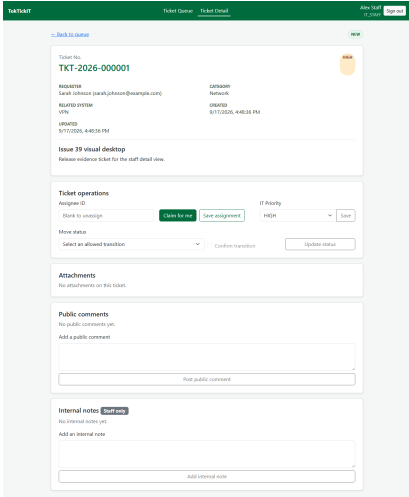
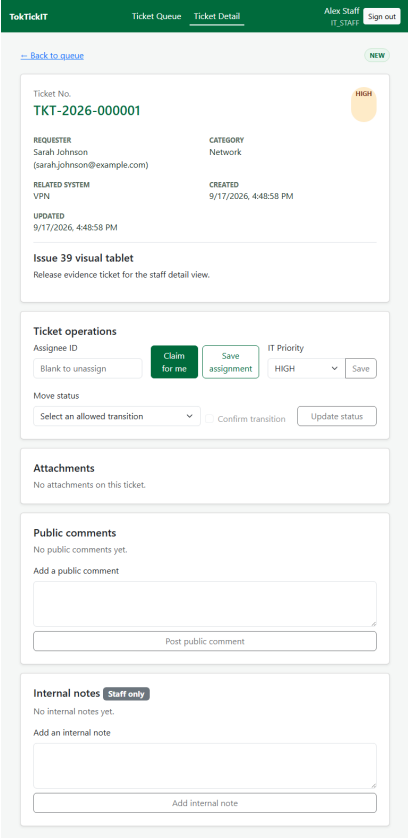
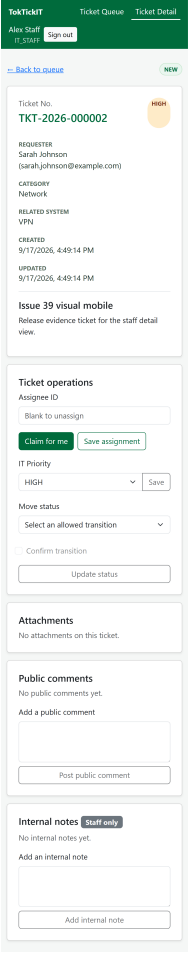
Mobile

The mobile card layout includes visible sorting controls, while desktop and tablet views show the table and filter hierarchy. The Open Detail action is backed by the staff detail route and regression coverage.

Supporting tests: server/tests/lab-03/staff-queue.api.test.ts; client/tests/lab-03/StaffTicketQueue.test.tsx; e2e/lab-03/staff-ticket-flow.spec.ts.

Answer Part 7: Working IT Staff Ticket Detail UI (10 Points)

Staff Ticket Detail supports claim/reassign, IT Priority, permitted status transitions and confirmations, attachment continuity, Public Comments, Internal Notes, Requester resolution indication, role restrictions, validation, and safe failure behavior.

IT Staff Ticket Detail responsive evidence		
		
Desktop	Tablet	Mobile

The three real-run captures show the ticket summary, operations, attachments, Public Comments, and staff-only Internal Notes at desktop, tablet, and mobile widths.

Supporting tests: server/tests/lab-03/staff-queue.api.test.ts and comments-notes.api.test.ts; client/tests/lab-03/StaffTicketDetail.test.tsx; e2e/lab-03/staff-ticket-flow.spec.ts.

The Administrator-only screen lists users, supports name/email search and role filtering, creates and edits one-role accounts, changes activation state, sets an initial password, and enforces duplicate-email and last-Administrator protections.

The image displays three mockups of a web application interface for 'TokTicket', specifically the 'User Management' section. The interface is designed to manage roles and account activation without exposing credentials.

Desktop View: The desktop layout features a top navigation bar with 'TokTicket', 'User Management', 'Ticket Queue', and user information ('TokTicket! Administrator', 'ADMIN', 'Sign out'). The main content area is titled 'User Management' and includes a sub-header 'Manage roles and account activation without exposing credentials.' Below this is a search bar with 'Search users' and 'Name or email' input, a 'Role' dropdown set to 'All roles', and a 'Create user' button. The user list is presented in a table with columns: Display name, Email, Role, State, and Actions. The table lists 11 users, including Alex Staff, Casey Staff, David Lee, Inactive Staff, Jamie Staff, Jennifer Anderson, Michael Brown, Robert Taylor, Sarah Johnson, and TokTicket Administrator. Each user row has a corresponding 'Actions' column with buttons for 'Edit', 'Deactivate', 'Reset password', and 'Activate' (where applicable).

Tablet View: The tablet layout maintains the same top navigation bar. The search bar is positioned below the header. The user list is displayed in a list format, with each user entry showing their name, email, role, and state, followed by a set of action buttons (Edit, Deactivate, Reset password, Activate) on the right.

Mobile View: The mobile layout also features the same top navigation bar. The search bar is located below the header. The user list is presented in a list format, with each user entry showing their name, email, role, and state, followed by a set of action buttons (Edit, Deactivate, Reset password, Activate) on the right.

Supporting tests: `server/tests/lab-03/users-admin.api.test.ts` and `authorization.api.test.ts`; `client/tests/lab-03/UserManagement.test.tsx`; `e2e/lab-03/user-administration.spec.ts`.

Answer Part 9: Zen Green UI and Responsive Evidence (5 Points)

The released screens reuse the Zen Green tokens and shared controls established in Lab 2. Visual tests and Playwright projects cover desktop, tablet, and mobile layouts for authentication, User Management, Staff Queue, and Staff Ticket Detail.

View	Desktop	Tablet	Mobile
Authentication	Login hierarchy, password visibility, safe errors	Scaled form with stable spacing	Single-column form without overflow
User Management	Table/list actions and filters	Readable list with preserved actions	Responsive cards and filter controls
Staff Queue	Table, sortable columns, pagination	Filter layout remains usable	Sortable cards with visible controls
Staff Ticket Detail	Operations and communications panels	Panels stack without clipping	Readable stacked controls and notes

Completed visual checklist

- Zen Green colors, typography, field styling, and button hierarchy are consistent across role shells.
- Loading, empty/no-results, validation, failure/retry, and submitting states are represented.
- Focus order, labels, safe error messages, readable badges, and editable/read-only boundaries are checked.
- No clipping, overlap, horizontal overflow, or hidden mobile queue sorting controls remain.

Submission status: contract, tests, AI-use record, reviewer history, responsive evidence, and final release are complete. Final Release PR #46 is merged into main at 92fac36; this report is the concise PDF required by the Lab 3 handout.

Lab 3 exclusions: external identity providers, password-recovery email, multi-role accounts, staff assignment history, and administrative deletion remain outside the approved scope. Deactivation is used instead of deletion so historical ownership and authored communication remain queryable.